

Tomáš Trachta

C# .NET Developer — high-performance tools & APIs

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ABOUT ME

.NET developer designing and building lower-level tools as well as HTTP APIs.

Most recently I built a Bash-compatible shell for Windows from scratch — 24,000 lines of dependency-free C++17.

I am mainly interested in libraries, tools and languages that enable other programmers to actually enjoy their craft.

I have a strong focus on reliability and performance, and I care deeply about the increasingly rare craft of well-optimized software.

CORE SKILLS

- **Languages:** C# .NET, C++17 (MSVC), Win32/COM, Python 3
- **.NET & concurrency:** async/await, TPL, System.Threading, WPF/MVVM
- **Testing:** xUnit / NUnit, FluentAssertions, Testcontainers, golden-file tests
- **Web development:** TypeScript, Vue, T-SQL, Docker, ASP.NET Core
- **Practices:** API design, performance optimization, open source

PROFESSIONAL EXPERIENCE

FinForce — .NET Developer

Apr 2025 – Present · Prague

Financial and internal processing systems for the Investona platform built in C# .NET and Kotlin.

STYRAX, a.s. — .NET Developer

Apr 2024 – Mar 2025 · Prague

Enterprise systems for health insurance companies.

MEMOS Software — Full-Stack Developer

May 2023 – Apr 2024 · Prague

Custom software solutions ranging from smaller customer applications to a full-scale CRM system for a multinational corporation. All built in C# .NET and Vue.js.

PROJECTS

All open source and built for my own daily use, because existing tools didn't satisfy me.

Joshua — dual-pane file manager for Windows

C#, .NET 9, WPF

github.com/tomas-trachta/Josha

- **Async architecture end to end:** `IFileSystemProvider` abstraction over local and remote (FTP/FTPS/SFTP) file systems, file-operation queue on `System.Threading.Channels`, connection pool with `IAsyncDisposable` lease semantics
- **Snapshot-backed disk-usage analyzer** with depth-bounded parallel scanning, custom virtualized canvas tree renderer, and DPAPI-encrypted, failure-safe persistence

wbsh — Bash-compatible shell for Windows

C++17, Win32

github.com/tomas-trachta/wbsh

- **Complete shell pipeline from scratch:** POSIX lexer → recursive-descent parser → arena-allocated AST → expansion engine → executor with job control
- **~35 coreutils and 40+ builtins**, including a self-contained AWK interpreter and DEFLATE decoder
- **Raw-mode line editor** (history, reverse search, limited per-tool tab completion); released with installer

Switchr — GPU-accelerated window switcher with virtual workspaces

C++17, Direct2D/DirectComposition

github.com/tomas-trachta/Switchr

- Overlay switcher on a D3D11/DirectComposition/Direct2D stack with **GPU-driven animations**
- **Alt-Tab-correct window enumeration**, MRU ordering via WinEvent hooks, relevance-scored search, and virtual workspaces with batched `DeferWindowPos` updates

EDUCATION

Charles University, Prague

Oct 2021 – May 2023 (unfinished)

Computer Science